

Support Vector Machine

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We consider binary classification with training data

$$S = \{(\mathbf{x}_i, y_i)\}_{i=1}^m, \quad \mathbf{x}_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}.$$

A linear classifier has the form

$$h_{\mathbf{w},b}(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b),$$

with decision boundary

$$H_{\mathbf{w},b} = \left\{ \mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0 \right\}.$$

When the data are linearly separable, many different hyperplanes may achieve zero training error. Support vector machines choose among these separators by maximizing the distance between the decision boundary and the closest training examples.

1 Margin and Hard-SVM

We first compute the distance from a point $\mathbf{x}$ to the hyperplane $H_{\mathbf{w},b}$. Consider

$$\min_{\mathbf{z}} \frac{1}{2} \|\mathbf{z} - \mathbf{x}\|^2 \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{z} + b = 0.$$

The distance is (left as a HW problem)

$$\text{dist}(\mathbf{x}, H_{\mathbf{w},b}) = \frac{|\mathbf{w}^\top \mathbf{x} + b|}{\|\mathbf{w}\|}.$$

For a correctly classified training example, $y_i(\mathbf{w}^\top \mathbf{x}_i + b) > 0$, so its signed distance to the decision boundary is

$$\frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}.$$

The margin of the classifier on the training set is therefore

$$\gamma(\mathbf{w}, b) = \min_i \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}.$$

The maximum-margin classifier solves

$$\max_{\mathbf{w}, b} \min_i \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}.$$

The ratio above is invariant under the rescaling $(\mathbf{w}, b) \mapsto (c\mathbf{w}, cb)$ for any $c > 0$. Let $\rho = \min_i y_i(\mathbf{w}^\top \mathbf{x}_i + b) > 0$. Rescaling by $1/\rho$ gives an equivalent separator satisfying

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad \text{for all } i.$$

Under this normalization, the geometric margin is $\gamma = 1/\|\mathbf{w}\|$. Thus, maximizing the margin is equivalent to minimizing $\|\mathbf{w}\|$, giving the *hard-margin SVM*:

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{subject to} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, \dots, m. \end{aligned} \tag{1}$$

The objective is convex and the constraints are affine, so this is a convex quadratic optimization problem. Geometrically, the two hyperplanes $\mathbf{w}^\top \mathbf{x} + b = 1$ and $\mathbf{w}^\top \mathbf{x} + b = -1$ bound the margin region. Their distance is $2/\|\mathbf{w}\|$. Thus, minimizing $\|\mathbf{w}\|$ maximizes the width of the margin.

2 Soft-SVM and Regularization

Hard-SVM requires the training set to be linearly separable. To allow violations of the margin constraints, we introduce nonnegative slack variables $\xi_1, \dots, \xi_m \geq 0$ and replace $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1$ by

$$y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1 - \xi_i.$$

The variable ξ_i measures the amount by which the i -th margin constraint is violated. Soft-SVM balances the norm of $\mathbf{w}$, which controls the margin, with the average amount of constraint violation. For a parameter $\lambda > 0$, Soft-SVM solves

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \lambda \|\mathbf{w}\|^2 + \frac{1}{m} \sum_{i=1}^m \xi_i \\ \text{subject to} \quad & y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 1 - \xi_i, \quad i = 1, \dots, m, \\ & \xi_i \geq 0, \quad i = 1, \dots, m. \end{aligned}$$

The parameter λ controls the tradeoff between the two terms. A larger λ places more emphasis on keeping $\|\mathbf{w}\|$ small, while a smaller λ places more emphasis on fitting the training data.

For fixed $(\mathbf{w}, b)$, the constraints on ξ_i are

$$\xi_i \geq 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \quad \text{and} \quad \xi_i \geq 0.$$

Since the objective is increasing in each ξ_i , its optimal value is

$$\xi_i = \max \{0, 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b)\}.$$

Define the *hinge loss* for a margin value $z \in \mathbb{R}$ as

$$\ell^{\text{hinge}}(z) = \max\{0, 1 - z\} = \begin{cases} 1 - z, & z < 1, \\ 0, & z \geq 1. \end{cases}$$

Thus, if $z < 0$, the example is misclassified and incurs loss larger than 1. If $0 \leq z < 1$, the example is correctly classified but lies inside the margin, so it still incurs positive loss. Once $z \geq 1$, the example is correctly classified with sufficient margin and the hinge loss is zero.

For the linear classifier

$$f_{\mathbf{w},b}(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle + b,$$

the margin on an example $(\mathbf{x}, y)$ is

$$z = y f_{\mathbf{w},b}(\mathbf{x}) = y(\langle \mathbf{w}, \mathbf{x} \rangle + b).$$

Therefore, the hinge loss used in SVM is

$$\ell^{\text{hinge}}((\mathbf{w}, b), (\mathbf{x}, y)) = \max\{0, 1 - y(\langle \mathbf{w}, \mathbf{x} \rangle + b)\}.$$

Therefore,

$$\xi_i = \ell^{\text{hinge}}((\mathbf{w}, b), (\mathbf{x}_i, y_i)).$$

Eliminating the slack variables, Soft-SVM is equivalently

$$\min_{\mathbf{w}, b} \left\{ \lambda \|\mathbf{w}\|^2 + \frac{1}{m} \sum_{i=1}^m \ell^{\text{hinge}}((\mathbf{w}, b), (\mathbf{x}_i, y_i)) \right\}.$$

Thus, Soft-SVM can be viewed as regularized empirical risk minimization with the hinge loss.

The hinge loss is a convex upper bound on the binary classification loss. Indeed,

$$\mathbf{1}\{y(\langle \mathbf{w}, \mathbf{x} \rangle + b) \leq 0\} \leq \max\{0, 1 - y(\langle \mathbf{w}, \mathbf{x} \rangle + b)\}.$$

3 Optimality Conditions and Support Vectors

We now examine the optimality conditions of Hard-SVM (1). Introduce a multiplier $\alpha_i \geq 0$ for each constraint. The Lagrangian is

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^n \alpha_i \left[1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b) \right]. \quad (2)$$

Since the hard-margin problem is convex and, for linearly separable data, a strictly feasible separator can be obtained by rescaling, the KKT conditions characterize an optimal solution.

Stationarity with respect to $\mathbf{w}$ gives

$$\nabla_{\mathbf{w}} \mathcal{L} = \mathbf{w} - \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i = 0,$$

so

$$\mathbf{w} = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i. \quad (3)$$

Stationarity with respect to b gives

$$\sum_{i=1}^n \alpha_i y_i = 0. \quad (4)$$

The remaining KKT conditions are

$$\alpha_i \geq 0, \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1,$$

and complementary slackness:

$$\alpha_i \left[1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b) \right] = 0.$$

If a point lies strictly outside the margin, i.e., $y_i(\mathbf{w}^\top \mathbf{x}_i + b) > 1$, then complementary slackness implies $\alpha_i = 0$. Therefore, only examples satisfying $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$ can have nonzero coefficients in the representation of $\mathbf{w}$. These active training examples are called *support vectors*. In particular,

$$\mathbf{w} \in \text{span}\{\mathbf{x}_i : y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1\}.$$

This gives the precise meaning of the term “support vector machine”: the separating hyperplane is determined by training examples that support the margin.

4 Duality

The same Lagrangian also leads to a dual formulation of Hard-SVM. The dual function is obtained by minimizing the Lagrangian over the primal variables:

$$g(\boldsymbol{\alpha}) = \inf_{\mathbf{w}, b} \mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}).$$

Substituting (3) and (4) into the Lagrangian (2),

$$\begin{aligned} g(\boldsymbol{\alpha}) &= \sum_{i=1}^n \alpha_i + \frac{1}{2} \left\| \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i \right\|^2 - \left\| \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i \right\|^2 \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \left\| \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i \right\|^2 = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j. \end{aligned}$$

Therefore, the dual problem is

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \\ \text{subject to} \quad & \alpha_i \geq 0, \quad i = 1, \dots, n, \\ & \sum_{i=1}^n \alpha_i y_i = 0. \end{aligned}$$

Weak duality always gives dual optimum $\leq$ primal optimum. For hard-SVM with linearly separable data, Slater's condition holds, so strong duality gives equality: dual optimum = primal optimum.

Once the dual solution $\boldsymbol{\alpha}^*$ is found, the primal weight vector is recovered from

$$\mathbf{w}^* = \sum_{i=1}^n \alpha_i^* y_i \mathbf{x}_i.$$

For any support vector with $\alpha_i^* > 0$, complementary slackness gives

$$y_i \left((\mathbf{w}^*)^\top \mathbf{x}_i + b^* \right) = 1,$$

which can be used to recover b^* .

An important consequence is that the dual optimization problem depends on the data only through inner products $\mathbf{x}_i^\top \mathbf{x}_j$. This observation will later allow us to replace inner products by kernel functions and construct nonlinear classifiers.